

Nanyang JC 2018 H2 Chemistry Prelim Practical Exam**Apparatus**

2 × 50 cm³ burettes
1 × burette clamp
1 × stand
2 × funnel for filling burettes
1 × 250 cm³ conical flask
1 × 25 cm³ pipette
1 × pipette filler
1 × white tile
1 × 250 cm³ beaker, labelled “for waste”
1 × 100 cm³ beaker
1 × metal calorimeter
1 × stopwatch
1 × −10 °C to +110 °C at 1 °C thermometer
50 cm³ measuring cylinder
1 × glass rod
1 × Bunsen burner
1 × lighter
1 × tripod
1 × heatproof mat
1 × test-tube rack
1 × test-tube holder
1 × metal spatula
4 × test-tubes
2 × boiling tubes
3 × droppers
1 × wash bottle containing distilled water
Access to paper towels
Access to gas supply
Access to delivery tubes
Access to a clear plastic bag containing 3 × red and 3 × blue litmus paper, 2 × strips of filter paper, splints

Chemicals

label	per candidate	identity	notes
FB 1	200	solution containing a mixture of $0.119 \text{ mol dm}^{-3}$ of ethanedioic acid and $0.0149 \text{ mol dm}^{-3}$ of sodium ethanedioate	Dissolve 15.00 g of $\text{C}_2\text{O}_4\text{H}_2 \cdot 2\text{H}_2\text{O}$ & 2.00 g of $\text{C}_2\text{O}_4\text{Na}_2$ in each dm^3 of solution
FB 2	100	0.35 mol dm^{-3} sodium hydroxide	Dissolve 14.00 g of NaOH in each dm^3 of solution
FB 3	250 cm^3	aqueous sodium carbonate	Dissolve 8.00 g of aqueous sodium carbonate in each dm^3 of solution
thymolphthalein indicator	5 cm^3	thymolphthalein indicator	freshly-prepared
FB 4	5 cm^3	0.1 mol dm^{-3} of aqueous zinc chloride solution	Dissolve 13.64 g of ZnCl_2 in each dm^3 of solution
FB 5	$\sim 3 \text{ g}$	sodium nitrate	Weigh about 3 g of NaNO_3 in a boiling tube and seal the boiling tube

Bench Reagents

- aluminium foil
- aqueous potassium iodide
- aqueous silver nitrate
- aqueous potassium manganate(VII)
- dilute sulfuric acid
- aqueous sodium hydroxide
- aqueous ammonia
- limewater